Rocket Engine Analysis

Jack Curtis and Chris West
300 N College St, Physics & Astronomy,
Carleton College, Northfield, Minnesota, 55057
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Evolved from early concepts during ancient times, the modern rocket engine was pioneered by Konstantin Tsiolkovsky, Robert Goddard, and Hermann Oberth. The modern rocket engine is comprised of the combustion chamber, the injector, the nozzle, the exhaust cone and the propellant feed system. The engine converts chemical energy to kinetic energy to produce thrust. [5] Maximum efficiency is determined by the compression ratio of in the engine, the maximum power output is determined by the temperature gradients and the thrust is determined by velocity of the air upon exit of the engine. These quantities are derived through thermodynamic and mechanical analysis of the Brayton cycle. Engineering can maximize these quantities through manipulating each step in this cycle, by minimizing the heat loss, entropy creation and air flow imperfections.

I. COMPOSITION OF A ROCKET ENGINE

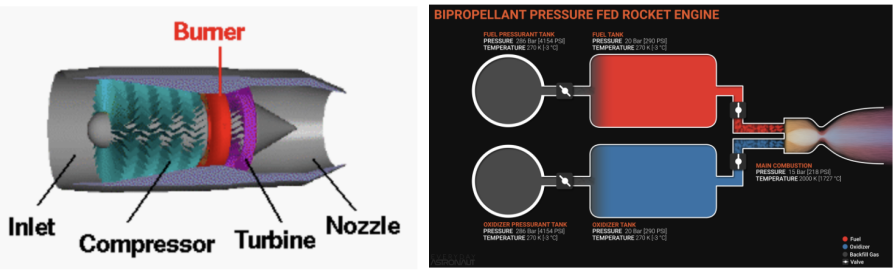


FIG. 1. Figure 1a shown on the left shows a simple cross section of a rocket engine. Figure 1b shown on the right shows a schematic of a bipropllant pressure fed combustion chamber.

The rocket engine's purpose is to move an aircraft or spacecraft through the air, necessitating thrust generation, also known as propulsion. Different than the Otto cycle which directs work to the drive chain. The most widely known and used form of propulsion is the gas turbine engine comprised a same key elements shown in figure 1a. [2]

The first main component is the compressor which compresses air intake from the inlet before the air passes into the combustion chamber. There are two main types of compressors, axial and centrifugal. An axial compressor takes on air parallel to the axis of rotation, while an centrifugal

compressor takes on air perpendicular to the axis of rotation. Most rocket engines use axial compressors because they can be linked together in order to form multistage axial compressors which can achieve higher compression ratios than centrifugal compressors which cannot be linked. The compression ratio defines the overall efficiency of the engine itself, which is derived in the next section. [2]

The next component is the combustion chamber. This component is where the propellant feed systems, shown in figure 1b, mix the fuel with the high pressure air and the burner applies heat to the gas. As the gas expands it does work on a power turbine which diverts some of the work back into the compressor. The remaining work is available as output to generate thrust. Figure 2 follows the cycle and demonstrates the work put back into the compressor as well as the work output.

The final components through which the gas passes is the nozzle and the exhaust cone, which is simply a specifically designed tube through which the exhaust gasses pass converts the air steam back to free flow, congruent with its surroundings. [3] [2]

The main differences between this system, which is governed by the Brayton cycle, and the internal combustion engine, which is governed by the Otto cycle, is that a rocket engine is a continuous combustion device in which each component of the engine serves one thermodynamic function. This is obviously much different from the intermittent combustion cycles present in the internal combustion engine, in which all thermodynamic processes occur in a piston–cylinder device. Despite this the thermodynamic processes are similar and can be analyzed the in the same manner.

II. IDEAL BRAYTON CYCLE

Studying this process thermodynamically, each part of this composition serves a thermodynamic function. In the first step the compressor, compresses the air adiabatically meaning there is no net heat flow or entropy change, $\Delta S = 0$ and $\Delta Q = 0$. In the combustion process, the gas is heated at a constant pressure, which is defined as an isobaric process. Following that, the gas expands adiabatically through the turbine and into the nozzle. Finally the gas expels heat through an isobaric process in the exhaust cone.

Efficiency is given as,

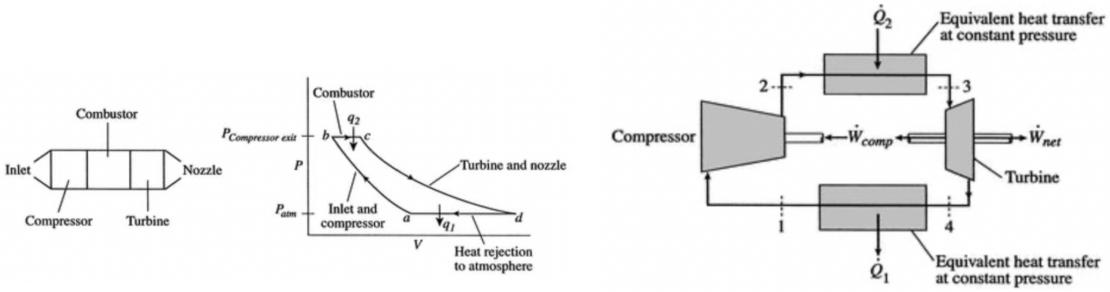


FIG. 2. Figure 2a (left) is a simplified jet engine as well as the PV diagram for the Brayton cycle with labeled stages. Figure 2b (right) is the path through which the air is passed and work is done.

$$\eta = 1 - \frac{benefit}{cost} = 1 - \frac{Q_c}{Q_h} \tag{1}$$

Mathematically the same as compared to the Otto cycle but Q_h is the heat input from the burner and Q_c is the heat expelled through the exhaust cone. The heats are defined as,

$$Q_h = c_p(T_3 - T_2) \rightarrow Q_c = c_p(T_1 - T_4) \tag{2}$$

where c_p is the specific heat at constant pressure. Taking their ratio;

$$\frac{Q_c}{Q_h} = 1 - \frac{(T_4 - T_1)}{T_3 - T_2)} \tag{3}$$

In order to simplify this equation notice that since the compression and expansion are ideally isentropic and therefore adiabatic from steps 1-2 and 4-1.

$$\frac{T_2}{T_1} = (\frac{P_2}{P_1})^{(\frac{\gamma-1}{\gamma})} = \frac{T_4}{T_3} = (\frac{P_4}{P_3})^{(\frac{\gamma-1}{\gamma})} = (\frac{P_1}{P_2})^{(\frac{\gamma-1}{\gamma})} \tag{4}$$

Therefore plugging in for T_4/T_3 and T_2/T_1 the expression simplifies as the following,

$$\eta = 1 - \frac{(\frac{T_3}{T_2} - 1)}{(\frac{T_2}{T_1})(\frac{T_3}{T_2} - 1)} = 1 - \frac{T_1}{T_2} = 1 - \frac{1}{(\frac{P_1}{P_2})^{(\frac{\gamma-1}{\gamma})}} \tag{5}$$

This result is identical to the efficiency of the otto cycle. In fact it is derived using the same thermodynamic processes present in both cycles. Despite the composition of the two engines and all of the physical differences present. The factor that governs the performance of the ideal engine is pressure ratio, also known as the compression ratio.

III. THRUST OUTPUT

Even though the efficiency derivation may be the same as the internal combustion engine, the purpose of the rocket engine is to generate thrust not to provide power to a drive train. Deriving a force equation from the thermodynamic and Newton's laws by using $F = ma$ and noticing that the momentum of the gases at the inlet and at the exhaust cone are related, and that the pressure over the engine area is a driving force, [6]

$$F = \dot{m}_4 V_4 - \dot{m}_1 V_1 + (P_4 - P_1)A_E, \tag{6}$$

where $\dot{m}$ is the mass flow rate of air through the engine, V is the velocity of the air per unit mass, and A_E is the cross sectional area of the engine.

In the ideal brayton cycle, the nozzle returns the gas to the atmospheric pressure such that $P_4 = P_1$. Therefore the third term goes to zero and defining the fuel to mass flow ratio as, $f = \dot{m}_{fuel}/\dot{m}_{air}$, the mass flow rates can be related by,

$$\dot{m}_4 = (1 + f)\dot{m}_1. \tag{7}$$

Therefore the force of the engines thrust is given as,

$$F_s = (1 + f)V_4 - V_1. \tag{8}$$

To solve for V_4 since V_1 is the velocity of the air in the atmosphere surrounding the engine in terms of the thermodynamic variables and efficiency,

$$\frac{T_4}{T_3} = (\frac{P_4}{P_3})^{(\frac{\gamma-1}{\gamma})} = 1, \tag{9}$$

which is the adiabatic relationships between the stages, the enthalpy equation ,

$$H = c_p T \tag{10}$$

and,

$$H_{Total} = H_4 - \frac{V_4^2}{2\eta_n} \tag{11}$$

where η_n is the efficiency of the nozzle, which is a result of the engineering design of the nozzle. Solving for V_4,

$$V_4 = \sqrt{2c_p T_4 \eta_n [1 - \frac{1}{(\frac{P_4}{P_3})^{\frac{\gamma-1}{\gamma}}}]} \tag{12}$$

The last term in the velocity is simply the efficiency of the brayton cycle which is dependent on the pressure/compression ratio from the previous section.

IV. POWER CONSIDERATION

The final quantity that is useful to know is the power output of the engine, specifically the maximum power output of the engine. For maximum efficiency the compression ratio is as high as possible. This means that the compressor exit temperature approaches the turbine entry temperature. The net work will be less than the heat received; as $T_2 \to T_3$ the heat received approaches zero and so does the net work. [6] A more useful quantity is the maximum output per unit mass that flows through the engine. This is given by,

$$\frac{W_{net}}{m} = Q_h - Q_c = c_p[(T_3 - T_2) - (T_4 - T_1)] \tag{13}$$

Investigating the temperatures in the system, T_3 is the max turbine temperature allowed, or the temperature limit at which the turbine blades melt/fail. This is a constant value as it is determined by the material used. T_1 is the atmospheric temperature and outside of the systems control, while T_2 is the compressor exit temperature which is controlled by the compressor system, which is variable due to the use of a multistage-axial compressor. Differentiating by this value,

$$\frac{dWork}{dT_2} = c_p[\frac{dT_3}{dT_2} - 1 - \frac{dT_4}{dT_2} - \frac{dT_1}{dT_2}] = 0 = -1 - \frac{dT_4}{dT_2} \tag{14}$$

therefore,

$$\frac{T_4}{T_1} = \frac{T_3}{T_2} \to \frac{dT_4}{dT_2} = -\frac{T_1 T_3}{T_2^2} \tag{15}$$

plugging this into equation 14 for dT_4/dT_2, $T_2 = \sqrt{T_1 T_3}$ therefore the maximum temperature for the compressor is

$$\frac{T_2}{T_1} = \sqrt{\frac{T_3}{T_1}}. \tag{16}$$

Plugging in equations 15, 16, and $T_2 = \sqrt{T_1 T_3}$ into equation 13 the maximum work output of a rocket engine is solely dependent on the heat capacity at constant pressure, c_p, and the temperatures throughout the cycle. Namely,

$$\frac{W_{net}}{m} = c_p[T_3 - \sqrt{T_1 T_3} - \frac{T_1 T_3}{\sqrt{T_1 T_3}} + T_1] = c_p T_1[\frac{T_3}{T_1} - 2\sqrt{\frac{T_3}{T_1}} + 1]. \tag{17}$$

In order to solve for pressure multiply by the mass flow rate per unit mass, $\dot{m}$ to find that the power output of a rocket engine is given by,

$$P = \dot{m} c_p T_1[\frac{T_3}{T_1} - 2\sqrt{\frac{T_3}{T_1}} + 1]. \tag{18}$$

V. THE NON IDEAL CASE AND ENGINEERING CONSIDERATIONS

The differences in the real efficiency of the brayton cycle is where the discrepancies from the Otto cycle become apparent. The Brayton cycle is generally more efficient in real-world applications compared to the Otto cycle because it operates at higher temperatures and pressures, which enhances thermal efficiency. In the Brayton cycle, continuous combustion allows for sustained high temperatures throughout the cycle, whereas the Otto cycle uses intermittent combustion in a closed chamber, which limits its temperature potential. Additionally, the Brayton cycle's continuous compression process is inherently more efficient than the rapid, single-stroke compression in the Otto cycle. These differences make the Brayton cycle ideal for applications like gas turbine engines in aircraft and power generation, where efficiency at high temperatures and continuous operation are essential, whereas the Otto cycle is more commonly used in traditional gasoline automotive engines.

The Brayton cycle, even though it reaches a higher efficiency than the Otto cycle, it is not ideal. This is because entropy is created due to high speed air flow through the engine (higher energy molecules) and thus is not an isentropic process. Another imperfection is that not all of the air flow goes through the compressor some of the airflow is diverted around the compressor and thus the engine is less efficient. Finally not all of the heat is transferred directly through the engine as it is engineered. Heat is lost to environment as well. The walls of the engine will allow some heat to escape as no material is a perfect insulator. Heat is also lost to the turbine and to the materials that compose the system itself. [4]

To solve these problems and maximize rocket engine efficiency within the modified Brayton cycle, engineers introduce several components and processes that optimize thermodynamic performance that minimize the entropy creation and heat loss. One such enhancement is the use of intercooling, where the compressed gasses are cooled between stages of compression. By reducing the temperature of the gasses before further compression, intercooling reduces the amount of work required by the compressors, which improves overall efficiency. This process allows the rocket engine to maintain high compression ratios without causing excessive heating that could reduce efficiency or damage components.

Reheating is another process employed to improve thermal efficiency, especially in multi-stage engines. In this process, partially expanded gasses are reheated between expansion stages, which maintains higher temperatures throughout the cycle. By reheating the gasses before each stage of expansion, engineers can extract more work from the fuel, increasing thrust without

significantly increasing fuel consumption. This strategy is particularly useful in staged combustion cycles and helps push the engine's performance closer to the Brayton cycle's theoretical limits. These considerations effect's on the thermodynamic process is shown in figure 3.

Turbo pumps are another component in modern rocket engines and are crucial for feeding high-pressure fuel and oxidizers into the combustion chamber quickly for maximum power output. Advanced turbo pumps with optimized turbine designs minimize energy losses during propellant pressurization and delivery, making the combustion process more effective.

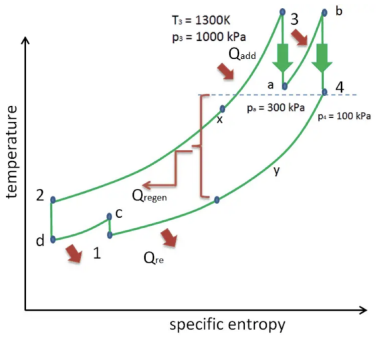


FIG. 3. A temperature and entropy diagram for a Brayton cycle with intercooling, heat regeneration, and reheating present.

Lastly, advanced nozzle designs play a pivotal role in converting thermal energy into kinetic energy. Innovations like bell nozzles and aero-spike nozzles enhance thrust by optimizing exhaust gas expansion across various altitudes. These advanced designs allow for higher exhaust velocities, maximizing the thrust generated per unit of fuel and enhancing the overall efficiency of the rocket engine. [7]

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